Middle Fork Nooksack Field Guide

A field guide to the land, water, rock, history, and living conditions of the Middle Fork Nooksack River corridor — what you are looking at when you drive, ski, climb, or work these roads.

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Access, Seasons & Safety

This is private working timberland, not a park. Plan accordingly.

Permit

- A **Hampton Recreation Permit** is required to enter. Permits are issued through hampton.myoutdooragent.com; see the Hampton Recreation Program for the current program and rules.
- The program is capped (roughly 600 motorized and 150 non-motorized permits per year), so buy early in the season.
- Gates use **red-painted locks**. A red lock is the signal you are on Hampton's permitted system.

Gates, roads & seasons

- The high country holds snow much of the year. The highest points around Bowman Mountain and Dailey Prairie (~3,200-4,000 ft) stay buried well into summer; weather windows, snow, and gate status drive every trip.
- Logging roads change constantly — re-grading, new spurs, washouts, and active landings. A road that went through last month may be gated, gone, or blocked by machinery.
- **Valley and airport forecasts often do not match the ridges.** Ridge microclimate (wind, temperature, snow cover, visibility) can be completely different from town. The southwest side of Bowman Ridge can behave very differently from the northeast side. See Live Conditions.

> **ACCESS ALERT (current):** December 2025 floods washed out **Forest Service Road 38 (Middle Fork Nooksack Road)**, cutting access to the **Elbow Lake** and **Ridley Creek** trailheads, and destroying the log crossing (with rope handrails) at the Elbow Lake trailhead. Repairs are planned but there was **no timeline** as of mid-2026. Verify road status before planning anything off FR 38. *(Source: Cascadia Daily News, Jun 16, 2026. Revisit this note as repairs progress.)*

Hazards

- **Active logging:** Crews, log trucks, road building, and rock crushing operate on these roads. Yield to industrial traffic, never block a road (you can be towed and fined), and stay out of active operations.
- **Active quarry:** The Olivine Corporation quarry on NW Twin Sisters is an active industrial site. Do not enter.
- **Old mine workings** are dangerous — observe from a distance.
- **Unstable slopes:** Watch for jack-strawed trees and DNR-mapped unstable ground (see Geology).
- **Glacier-fed rivers** rise through a warm day — a morning ford can be a serious crossing by afternoon.

Cultural respect

- The entire Middle Fork watershed is a **Traditional Cultural Property** of the Nooksack Indian Tribe. Treat cultural sites with respect; do not disturb cultural artifacts (also a condition of the Hampton permit).
- If you find something archaeological, **leave it in place**, photograph it with GPS, and report it to the Nooksack Tribe Cultural Resources Department or the Washington DAHP. Removing artifacts from state or federal land is illegal (RCW 27.53).
- See Traditional Cultural Property.

Nooksack Place Names

The Nooksack people (Lhéchalosem language) named this landscape long before any roads were built.

The entire Middle Fork watershed is a **Traditional Cultural Property** and a **Place of Significance** for the Nooksack Indian Tribe. It remains one of their primary areas for traditional religious activities because it is less disturbed than the North or South Forks.

Nooksack Name	English Name	Meaning
Nuxwt'iqw'em	Middle Fork Nooksack River	"always-murky water"
Kwetl'kwitl' Smanit	Twin Sisters Mountain	"red mountain"
Kweq' Smanit	Mount Baker	"white mountain"
Sheqsan / Ch'esqen	Mount Shuksan	"high foot" or "golden eagle"
Kwelshan	High slopes of Mt. Baker	"shooting place"

Reference: *Nooksack Place Names: Geography, Culture, and Language* by Allan Richardson and Brent Galloway (UBC Press, 2011). The Nooksack Tribe has published a place-name map through its Cultural Resources / Culture Program.

History of Place Names

Where the English names in this area come from — who these people were and why their names stuck.

Nooksack
From the Nooksack language word *Nuxwsa7aq*, the name of a village at the mouth of Anderson Creek (near present-day Goshen). It translates to "always bracken fern roots" — bracken fern roots were a dietary staple gathered there. Early settlers misheard it various ways ("Nooksahk," "Neuk-sack," "Buk-sak"). An alternative folk etymology, "mountain men," was reported by Edmund Fitzhugh in 1857 and appears in some older sources, but the fern-root translation is the one the Nooksack Tribe uses.

Galbraith (Mountain / Chromite Claims)
Named for the family of Audley A. Galbraith, who homesteaded 160 acres in Acme Township around 1884 with his wife Henrietta. A.A. Galbraith was killed in the 1911 Saxon-camp shootout (see [The Saxon Camp Murders]). Two of his sons, Joe and Hugh, became well-known loggers; **Joe Galbraith won the first Mt. Baker Marathon in 1911**. The "Galbraith Group" chromite claims on Twin Sisters (staked by J. Galbraith et al., August 25, 1934) are likely the same family. Galbraith Mountain near Bellingham (officially North Lookout Mountain) carries the family name too.

Bowman (Mountain)
The specific namesake is not definitively documented in published sources. The name appears in Whatcom County records by the early 1900s. Given the pattern of other local names (Galbraith, Deming, Van Zandt), it almost certainly honors an early settler or timber family who worked the area. The Nooksack Lookout (built 1949, demolished ~1968) sat on its 3,350-ft summit.

Deming
Named after George Deming, the community's first postmaster. Deming sits at the confluence of the Nooksack forks and has long been a hub for the Nooksack tribal community.

Van Zandt
Named after Julian Milner Van Zandt, who homesteaded in the area in 1883 and became a community postmaster. Dutch-origin surname.

Acme
Named around 1887. Two accounts survive, and both involve the **Park(s)** family and an "Acme" hymnal:

- Resident **Samuel Parks** named it after the Acme hymnal he used in church; it became the post office name in 1887.
- A variant account credits **George Parls**, who named it after a local church's newly received Acme hymn book.

Either way, the Greek word *akme* means "peak" or "highest point." *(See Acme, WA — Wikipedia and Revisiting Washington.)* Note the **Park(s)** family also appears in the naming of

Mosquito Lake (below), though by a different first name — likely the same pioneer family, but not confirmed as the same individual.

Saxon
Named for the Saxon logging camp operated by Bloedel-Donovan. The first logging camp in the area was established in 1905 just downstream of the Saxon Bridge. (Note: the 1911 murder victim Winn Stevens was superintendent of the separate **Key City Logging Company** camp near Saxon — see The Saxon Camp Murders.)

Maple Falls
Descriptive: named for the maple trees and the waterfall in the area. The town served as a railroad terminus for the Chicago, Milwaukee, St. Paul and Pacific Railroad, with a 35-mile rail haul to Bellingham.

Welcome
A small community on State Highway 542 at the turnoff to Mosquito Lake Road. The specific origin is not well-documented, but it follows the common 19th-century practice of choosing inviting names for new post-office communities.

Mosquito Lake
The lake (and road) take their name from the marshy, mosquito-heavy conditions around the lake. The name was applied by **Charles M. Park in 1885** for exactly the reason you'd guess. DNR Bulletin 57 (1969) documented sphagnum peat deposits at Mosquito Lake — the boggy, standing-water conditions that breed mosquitoes. The lake and Mosquito Lake Road are the main access corridor from the lowlands into the Middle Fork and Dailey Prairie area. *(Source: Mosquito Lake — Wikipedia.)*

Twin Sisters
A descriptive English name for the range's two prominent summits: North Twin (6,640 ft) and South Twin (~7,000 ft). The Nooksack name *Kwetl'kwitl' Smanit* ("red mountain") describes the oxidized dunite color rather than the twin-peak shape.

Dailey Prairie
Named for "Old Man Dailey" — a solitary figure who lived in a small cabin on the edge of the prairie. Little is known about him beyond local oral tradition passed down through loggers. The cabin survived into at least the 1990s (Dave Tucker photographed it in 1990) but was eventually burned down — reportedly by federal authorities, consistent with the common USFS practice of demolishing unauthorized backcountry structures. The prairie sits atop a broad pass (~3,200 ft) between Bowman Mountain and the Twin Sisters Range, with the north side draining into Galbraith Creek and the south into Orsino Creek (Skookum drainage). It is now a 229-acre DNR Natural Area Preserve. No "Dailey" appears in the published pioneer biographies of Whatcom County (Roth, 1926), in the GLO patent index, or in Edmond Meany's *Origin of Washington Geographic Names* (1923) — the name was likely applied informally by loggers

and forest users who knew the old man and his cabin, and it stuck on maps before anyone recorded who he was.

Bloedel-Donovan (Park, Lumber Mills)
Julius H. Bloedel (banker from the Midwest, moved to Fairhaven ~1890) and John J. Donovan (son of Irish immigrants, arrived in Fairhaven ~1889) partnered with railroad contractor Peter Larson to create the Lake Whatcom Logging Company in 1898. By 1913 their operation, Bloedel-Donovan Lumber Mills, was one of the largest on the West Coast — they owned an estimated 1.2 billion board feet of standing timber in Whatcom and Skagit Counties. They built 50+ miles of logging railroad through the Saxon-South Fork corridor (1920-1937). Donovan died in 1937; Bloedel donated the mill site to Bellingham as a park in 1946 and died in 1957.

Swen Larsen (Quarry)
The original name for the Olivine Corporation quarry on the NW end of Twin Sisters. Named for the individual who established early quarrying claims on the site. The quarry has been operated by the Olivine Corporation of Bellingham since 1963.

Skookum (Creek / Drainage)
From Chinook Jargon — the pidgin trade language used across the Pacific Northwest by both indigenous peoples and early settlers. *Skookum* means "strong," "powerful," "brave." Applied to creeks and rivers throughout the region to indicate powerful water. Orsino Creek drains south from Dailey Prairie into the Skookum drainage. The Nooksack name for Skookum Creek is *Nuxwaynaltxw*, meaning "Slaughterhouse, a place to dry, a place to prepare" — a salmon processing site.

Orsino (Creek)
A GNIS-registered stream (N48.69039 W122.04071, Cavanaugh Creek quad) draining south from the Dailey Prairie pass area at ~2,073 ft elevation into the Skookum drainage. The name *Orsino* is Italian, meaning "little bear" (from Latin *Ursus*), and is also the name of the Duke in Shakespeare's *Twelfth Night*. Whether the creek was named for an Italian settler/prospector or by someone with a literary bent is not documented. No "Orsino" appears in Whatcom County pioneer records. Like "Dailey," this may be a name applied informally by backcountry users and adopted by the USGS surveyor who mapped the Cavanaugh Creek quad.

Heislers (Creek)
A GNIS-registered stream (N48.78234 W122.11182) flowing into the Middle Fork Nooksack area near Mosquito Lake Road. The possessive form "Heislers" strongly suggests it was named for a specific individual or family surnamed Heisler — a German surname meaning "cottager" or "small house owner" (*hausler*). No Heisler appears in the published Whatcom County biographies (Roth, 1926), but the name could reference a logger, trapper, or small landowner. Whatcom County had significant German and Swiss-German settlement in the late 1800s.

Johnsons (Creek)
A GNIS-registered stream (N48.75873 W122.11071) near the Middle Fork corridor. Not to be confused with the separate Johnson Creek near Kendall (N48.99 W122.25). The possessive form again points to a specific person. Historical records mention a Bill Johnson who built a square-notched log cabin in the Hopewell area and a Nooksack tribal member named David Johnson — but neither is confirmed as the namesake. Johnson is the most common surname in early Whatcom County records, making attribution difficult.

Peterson (Creek)
A GNIS-registered stream (N48.72206 W122.09904) in the Middle Fork area. Named for a person surnamed Peterson. Whatcom County had many Petersons — the Scandinavian immigration wave of the 1880s-1900s brought dozens of families with this name. The most documented is George S. Peterson, born in Minnesota in 1881, who arrived in Whatcom County in 1900, worked in lumber camps, and later established a shingle mill with his brothers. Whether this Peterson or another is the creek's namesake is not confirmed.

Seymour (Creek)
A GNIS-registered stream (N48.75762 W122.00987) flowing into the Middle Fork Nooksack near the Twin Sisters. Seymour is a Nooksack tribal family name. Historical and contemporary tribal records document individuals including Hamilton Seymour and Moses Malloway Seymour as Nooksack members. This is notable — most creek names in the corridor honor Euro-American settlers, but Seymour Creek likely honors a Nooksack family, reflecting the tribe's deep, ongoing presence in the Middle Fork watershed. The English surname Seymour (from the French village of St. Maur) was adopted by some indigenous families during the allotment and assimilation era.

Road Names Cheat Sheet

Road codes follow the pattern *(abbreviation)*-**ML**, where **ML** = "Mainline." Mainlines are named after **drainages**, so the prefix abbreviates a drainage name (e.g., **MF** = Middle Fork, **MM** = Musto Marsh).

Code	Full Name	Local / Map Name
MM-ML	Musto Marsh Mainline	—
MF-ML	Middle Fork Mainline	Middle Fork Road
WM-ML	*(unverified drainage)*	"Weyerhaeuser Road #9000" (legacy)*

* The original guide labeled **WM-ML** "Welcome Mainline," but that name can't be sourced and doesn't fit the drainage convention. The only mappable name in that row, **Weyerhaeuser Road #9000**, *is* a real legacy gravel road (OSM / Cartogiraffe) — but it lies up by the **Twin Sisters / Dailey Prairie** (near the Swen Larsen/Olivine quarry and Galbraith & Orsino creeks), *not* the community of Welcome. The drainage behind "WM" is still unidentified; confirm against a Hampton/DNR road map before trusting this row.

Geology: Things You Can Walk Up To

1. Lahar deposits on the Middle Fork riverbanks

The Middle Fork contains deposits from the **largest lahar ever to come off Mount Baker** (~6,500 years ago). It began with the collapse of the Roman Wall — the headwall of the Deming Glacier — and transformed into a concrete-like slurry that traveled all the way to the lowlands. A published GSA field-trip guide (Dave Tucker, 2014, GSA Field Guide 38, pp. 33-52) has GPS stops along the road.

What to find:

- **End of the Middle Fork road (near the old dam site):** Walk to the gravel bars. The moraine of an older valley glacier is capped by the Middle Fork lahar deposit. Look at terrace cuts in the riverbanks — the older, clay-rich lahar is a distinct gray layer.
- **Ridley Creek lahar exposure (farther downstream):** The Ridley Creek lahar sits directly on top of the Middle Fork lahar. The Ridley Creek layer has a distinctive orange tint and a fine clay matrix. You can put your hands on the boundary between two separate lahars.
- **Elbow Lake trailhead area (north bank):** A ~1,600-year-old debris-flow deposit mantles a 7-meter-tall terrace. Angular rock fragments in a gray, clay-poor matrix — this was likely a glacial outburst flood that bulked up into a debris flow, not a volcanic event.

Sources / photos: USGS lahar photo · MBVRC Middle Fork lahar field trip · NW Geology — Ridley Creek Trail geology guide

2. The 1927 Deming Glacier jokulhlaup (and the 2013 debris flow)

The same terraces hold evidence of two much more recent, more sudden events:

- **June 1927 — Deming Glacier outburst flood (jokulhlaup):** Eyewitnesses reported a giant flood carrying "house-sized blocks of ice and boulders" down the river, caused by the collapse of the stagnant lower ~1 km of the Deming Glacier. It left **garage-sized lava-breccia boulders** on the terraces, about 2.3 miles downstream from the current glacier terminus. Tree cores on the low terrace near the Ridley Creek Trail date the oldest trees to

1927, confirming the event. Deposits are recognized as far downstream as the Elbow Lake trail crossing.
- **May 31, 2013 — debris flow:** A more recent flow (most likely a landslide off a moraine above the Deming Glacier terminus) left a new **boulder levee about 15 ft above the channel**, with mud-coated andesite boulders up to 10 ft across and trees scarred up to ~6 ft. It was **detected on Pacific Northwest Seismic Network seismometers**, which raises the possibility of downstream warning for future large events.

Sources: NW Geology — Ridley Creek Trail guide · MBVRC — May 31, 2013 debris flow

3. Hunt for "Shermanite" in the river gravel

On the gravel bars, look for pebble- to cobble-sized pieces of **white to cream-yellow, crumbly, clay-rich rock**. This is "Shermanite" — the informal name geologists use for hydrothermally altered andesite from Sherman Crater on Mount Baker, carried here by lahars and floods. It looks bleached and wrong compared to everything else in the gravel. When you find it, you are holding rock that was once part of the active volcanic crater wall.

4. Twin Sisters dunite — a chunk of Earth's mantle

The Twin Sisters range (the jagged reddish-brown peaks to your southeast/east) is the **largest exposed body of dunite in the contiguous US** — a 16 km x 6 km slab of mantle rock, tectonically shoved to the surface.

- The reddish-brown color on the peaks is iron/magnesium oxidation (fresh dunite is green — olivine).
- Sparse vegetation on lower slopes because ultramafic soil is toxic to most plants (low phosphorus, heavy metals). The Nooksack people saw this and named it "red mountain" — *Kwetl'kwitl' Smanit*.
- Black chromite layers are visible as complexly folded bands within the dunite — these were the subject of WWII-era mining investigation (USGS Open-File Report 43-39, 1943) when the US government urgently needed domestic chromium.

Sources / photos: Washington Landscape — Twin Sisters dunite (Dan McShane) · WA DNR lidar/photo composite · Dunite + chromite close-up

5. Clay Banks Landslide (near Deming)

About 1.8 miles upstream of the Baker Highway bridge: an active, USGS-monitored landslide complex in glacial terrace material. In February 2014, it was large enough to temporarily divert the entire Nooksack River. The exposed bluffs of glacial sediment are visible from accessible vantage points. *(Monitoring: USGS Clay Banks East landslide site.)*

6. Jack-strawed trees (slope-instability indicator)

Throughout the corridor, watch for trees tilted and fallen in mixed, crisscross directions — not all leaning the same way (which would indicate wind). This "jack-strawed" pattern indicates the ground itself has moved (landslide, creep, or deep-seated instability). Mapped as "potentially unstable slopes" by DNR. Treat as a clue to stay alert.

7. Van Zandt Dike (visible from Highway 9)

Between Deming and Acme on Highway 9, you pass a large forested plateau rising abruptly from the valley floor on the east side of the road. This is the **Van Zandt Dike** — not a volcanic dike but a massive ancient bedrock landslide from the Chuckanut Formation, roughly 13,000 years old, that slid into the Nooksack Valley and dammed or displaced the river. It is one of the largest known landslide deposits in the county, and is visible (labeled) from the Nooksack Lookout summit. Dan McShane has written extensively about the Van Zandt Landslide Complex using lidar: washingtonlandscape.blogspot.com.

Dam Removal Site

City of Bellingham Diversion Dam (removed 2020)

Visible on maps as "City of Bellingham Diversion Dam" in the upper Middle Fork corridor. The 24-foot dam was built in 1961, lacked any fish passage, and blocked 16 miles of upstream spawning habitat.

- **Removed in 2020**, channel restoration completed 2022.
- **Collaboration:** City of Bellingham, Nooksack Tribe, Lummi Nation, American Rivers, NOAA, WDFW, Paul G. Allen Family Foundation.
- Spring Chinook, steelhead, and bull trout now access the upstream habitat for the first time in 60 years.
- The new water intake (with fish-compliant screens) is just upstream of the former dam site.

Status update (as of mid-2020s): All project elements were complete in 2022. **Effectiveness monitoring runs through 2030.** The Year-3 report (covering 2023, published spring 2024) found no channel changes that threaten fish passage, even after the major November 2021 floods. The project was the #1 Recovery Action of the WRIA 1 Salmonid Recovery Plan and is modeled as the **single largest Nooksack Chinook population gain** of any restoration action — projected ~31% abundance increase, ~12% productivity increase, ~48% diversity increase — and is tied to **Southern Resident Killer Whale recovery** through increased Chinook prey. *(Sources: City of Bellingham — project status and benefits; American Rivers.)*

How the water gets to Bellingham — and why they punched a tunnel through a mountain:
Lake Whatcom is Bellingham's drinking-water reservoir (100,000+ people). By the 1950s the city decided the lake alone wasn't enough for future growth. The Middle Fork Nooksack — 20 miles east, on the other side of Bowman Mountain — had plenty of clean water. Problem: Bowman Mountain is in the way, and the two are in different watersheds. The city's solution (completed 1961): build a dam on the Middle Fork, then bore a **1.6-mile tunnel (~8 ft diameter) straight through Bowman Mountain**. Water flows by gravity through the tunnel, then into a 9.5-mile pipeline to Mirror Lake, which drains into Anderson Creek and then Lake Whatcom. The whole system runs on gravity — no pumps. Water is diverted intermittently, mostly during winter and spring high flows. The new fish-screened intake (700 ft upstream of the old dam site) still feeds this same tunnel-and-pipeline system today. The city kept its water supply; the fish got their river back.

Galleries / video: City of Bellingham before/during/after photos · NOAA Fisheries feature · Time-lapse video

First Salmon and what the dam blocked

The dam didn't just block fish — it blocked a relationship. The Nooksack Tribe's **First Salmon ceremony** is one of the oldest continuing spiritual practices in the Pacific Northwest. When the first spring Chinook of the season is caught, a designated fisher prepares it according to strict protocol: the head is kept pointed upstream (to guide the spirit home), the community shares the fish, and the bones are cleaned and returned to the river — because the salmon will reconstitute itself and return next year, but only if it is treated with respect. No general fishing is permitted until the ceremony is complete. This reciprocal relationship was severed for 60 years when the dam cut off the Middle Fork spawning habitat. With the dam gone and Chinook returning upstream for the first time since 1961, the ceremony's meaning in this watershed is being restored along with the fish.

Mines & Mineral Workings

1. Olivine Corporation Quarry (active — do not enter)

- **Location:** SE 1/4 NE 1/4, Section 34, T38N R6E — NW end of Twin Sisters Mountain, ~1.25 miles south of the Middle Fork.
- **Access road:** From Mosquito Lake Road, turn onto the old Nooksack Lookout road, follow logging roads ~11 miles east.
- **What you see from the road:** A 200 x 700-foot stripped area. Weathered olivine on the surface is red/brown; fresh rock below is dark-gray to green. Silvery enstatite veinlets and

tiny black chromite grains visible in specimens.

- **History:** Originally the Swen Larsen Quarry. Olivine mined for refractory brick as early as 1946. Olivine Corporation began formal quarrying in 1963. Rock is trucked 40 miles to their Bellingham grinding plant (foundry sand, blast sand).
- The quarry is on Hampton land and is an **active industrial site — do not enter.** But the access road and margins have dunite road cuts and loose specimens.

2. Asbestos veinlets on Bowman Mountain

- **Location:** NW 1/4, Section 29, T38N R6E — about 4 miles NW of Twin Sisters, in the Bowman Hill logging area.
- **What you see:** Thin chrysotile (white asbestos) veinlets, typically under 1/4 inch wide, in serpentinized peridotite (green-black rock with silky white cross-fibers). One of only two locations in the county where veinlets reach noticeable width (DNR Bulletin 57).
- **Caution:** Do not break or crush serpentinite — asbestos fibers become airborne. Observe only. *(For reference, what chrysotile veins look like in serpentinite: Wikimedia photo.)*

Chromite & related claims (1920s-1930s):

Claim	Date	Details
Sister & Skookum (Skookum Creek headwaters)	Jul 1926	Earliest claims; M.M. and S.S. Lambert
Ribbon (Green Creek)	1930s	Open cut; ore hauled 1,200 ft down a wire tramline to a camp at 3,000 ft, then out by mule train
Danny (head of South Fork)	1930s	Chromite from lenticular veins in a cliff 800 ft above valley; ~3,000 lbs extracted
Good Hope	1930s	High-grade ore identified, limited development
Washington Chrome Co.	Jan 1931	W.M. Elsey et al.
Thunder Mountain Group	Jul 1931	A. Odmark et al.

3. Chromite claims on Twin Sisters (1930s)

Scattered across the upper mountain (not casually accessible). USGS concluded in 1943 that deposits were too small/scattered for commercial mining — individual pods typically under 1,000 tons each. The WWII strategic-minerals effort came, looked, and moved on.

4. Nooksack Lookout site on Bowman Mountain (tower demolished ~1968)

The Nooksack Lookout (also called Bowman Mountain Lookout) was on top of Bowman Mountain at 3,350 ft. Built in 1949 as an 80-foot pole tower with an L-6 cab and separate sleeping quarters. In 1955 and 1956 it was staffed by **Lorna Anderson** — a local woman and University of Washington student who spent two fire seasons living alone on the summit watching for smoke. In 1957 a DNR carpentry crew cut the tower to 50 ft and installed a 9x9 ft cab. The tower and ground cabin were destroyed around 1968. Possible footings, bolts, or cleared ground remain at the summit. The "Nooksack Lookout road" referenced in DNR Bulletin 57 leads toward this site and continues to the Olivine quarry.

Photos: 1956 tower standing (Fire Lookout Museum) · Surviving footings & summit views (WillhiteWeb)

Abandoned Quarries Worth Visiting

1. Marble Peaks / Boulder Creek limestone quarry

- **Location:** W 1/2 Section 22, T40N R6E — about 3 miles NE of Maple Falls, west bank of Boulder Creek.
- **What you see:** Abandoned water-filled quarry and impressive limestone outcrops ("Marble Peaks"). White to gray crystalline limestone containing large crinoid stem fossils (ancient sea-lily segments — look like stacked poker chips). Contains serpentine inclusions. Operated by Mitchell Bay Lime Co. until 1952. The last mile of access road is abandoned.

2. Doaks Creek limestone quarry

- **Location:** Near center of Section 19, T40N R6E — about 1.25 miles north of Maple Falls.
- **What you see:** Abandoned quarry with a 100-ft-long, 40-ft-high working face. Dark-gray limestone with fossil coral beds up to 3 ft thick (Devonian age — ~380 million years old). The coral makes the rock distinctly banded. Operated by Mitchell Bay Lime Co. 1953-1954.

Historical Logging & Homestead Remnants

1. Springboard-notch stumps

Throughout the second-growth forest, look for enormous old-growth stumps (6-10 ft diameter) with rectangular notches cut into the sides, typically 4-8 feet above ground. Loggers chopped these slots, inserted wooden planks (springboards) to stand on, and hand-sawed the tree above the thick butt swell. Bloedel-Donovan Lumber Mill (Lake Whatcom, founded 1901) drove the first-generation logging in this watershed. *(Photos: Bloedel-Donovan era logging (Wikimedia); turn-of-the-century logging, Whatcom County.)*

2. Steel cable and choker hardware

Rusted 3/4" to 1" braided steel cable from steam-donkey yarder operations. Found in coils, strung between stumps, or buried in creek bottoms. Bronze or steel choker bells (cable fittings) turn up in road cuts and stream beds.

3. Skid-road depressions

Parallel depressions in the forest floor where greased wooden skids were laid and logs were dragged over them by steam power. They appear as unnaturally level, curving paths, wider than a game trail.

4. Bloedel-Donovan railroad grades (Saxon area)

Between Acme and Saxon (South Fork), Bloedel-Donovan built 50+ miles of logging railroad from 1920-1937. The rails were pulled when the operation ended (~1940), and many grades were converted to truck roads. Look for railroad spikes and tie plates, concrete or timber bridge abutments, and trestle pilings in stream crossings (visible at low water).

5. The Saxon Camp Murders (1911)

On **January 5, 1911**, Audley A. Galbraith Sr. (age 60, Justice of the Peace, and the man Galbraith Mountain is named for) and **Winslow B. "Winn" Stevens** (age 45, superintendent/foreman of the **Key City Logging Company** camp near Saxon) confronted two transients who had stolen a shipment of logging boots from the company's freight delivery at the Saxon spur.

Stevens tracked the men to a shingling shed in Acme and asked Galbraith — who was unarmed — to arrest them. The confrontation became a gunfight. Galbraith was shot five times (reportedly by the man known only as "**Slim**") and died at the scene. Stevens, who carried a .32-caliber revolver and returned fire, was shot in the abdomen and died about a week later in Sedro-Woolley.

The two men fled by freight train to Sedro-Woolley. One, **Mike Donnelly** (wounded in the stomach by Stevens), was captured on January 6 after a doctor recognized his gunshot wound and called police. He was convicted and sentenced to life at Walla Walla. "Slim" was never found. The Galbraith family remained prominent in Whatcom County — their name is on the mountain, the creek, and the chromite claims, and son **Joe Galbraith won the first Mt. Baker Marathon later that same year** (see Recreation).

(Note: earlier drafts variously attributed Stevens to the "Saxon logging camp" or Bloedel-Donovan; the contemporary record places him with the Key City Logging Company camp. Sources: [HistoryLink — Mike Donnelly capture](); [WhatcomTalk — "What's in a Name?"]().)

6. McBride Homestead (Middle Fork floodplain)

The 1899/1900 GLO survey maps show a "McBride" residence on the lower Middle Fork floodplain. Signs of an old homestead: unnaturally flat clearings with out-of-place fruit trees (apple, cherry), foundation stones, and rusted wire fencing grown into tree trunks.

Flora: Rare & Notable Plants

1. Dailey Prairie Natural Area Preserve

Visible on maps at ~3,200 ft elevation, atop a broad pass between Bowman Mountain and the Twin Sisters Range. A **229-acre DNR Natural Area Preserve** — one of the highest-quality natural wetland systems remaining in Washington. Named for "Old Man Dailey," who kept a cabin at the prairie's edge (see History of Place Names).

- **Several-flowered sedge** (*Carex pluriflora*): State Sensitive plant. Largest known population in Washington.
- **Sphagnum bog:** A true sphagnum peatland — rare in the Cascades at this elevation.
- **Four-needle pines:** Local observers report unusual pines with four-needle clusters at the prairie — likely whitebark pine (*Pinus albicaulis*, a five-needle pine) at the extreme low end of its range, or an atypical variant. Worth investigating.
- **Other species:** bog laurel, marsh violet, woolly sedge, white marsh-marigold.
- Surrounded by old-growth silver fir, mountain hemlock, and Alaska yellow-cedar.
- North side drains into Galbraith Creek; south side drains into Orsino Creek (Skookum drainage).
- *Photos / report:* Chuckanut cross-country skiing — Dailey Prairie · crawford.tardigrade.net album

2. Twin Sisters Sandwort (*Sabulina sororia*)

A Washington State Endangered plant that exists only on Twin Sisters — endemic to a 16 km² patch of serpentine-like rocky slopes. Formally described as a new species in 2016 (previously misidentified as *Arenaria rossii*). You won't find it anywhere else on Earth. *(Photos: Burke Herbarium; species description: PMC.)*

3. Old-growth forest remnants

The Middle Fork corridor preserves pockets of old-growth forest in steep ravines and canyons:

- Pacific silver fir and western hemlock dominate.
- Alaska yellow-cedar at higher elevations.
- Multi-layered canopies, large-diameter trees, abundant coarse woody debris.
- Fire return interval exceeds 700-1,000 years in these forests — they are shaped by windthrow, not fire.

Fauna: What You Might See

1. North Cascades Elk Herd ("Nooksack Herd")

The northernmost elk herd in western Washington. A mix of Rocky Mountain and Roosevelt elk genetics. Re-established through augmentations in 1946-48 and 2003-05 (Mt. St. Helens stock). Core range is the South Fork and middle Skagit, but they move through the Middle Fork corridor in summer.

2. American Dipper (Water Ouzel)

Stocky, dark-gray bird standing on rocks mid-stream, bobbing up and down. *Cinclus mexicanus* — North America's only aquatic songbird. It walks underwater on the river bottom to hunt insect larvae. Commonly found along the Middle Fork. Its song carries over rapids. Look for them near fast water and waterfalls.

3. Fisher — a recovery story (reintroduced 2018-2020)

The **fisher** (*Pekania pennanti*) — a large, dark forest weasel once trapped out of Washington — has been brought back to the North Cascades. From 2018 to 2020, **89 fishers captured in Alberta were released across the North Cascades, including the Mount Baker-Snoqualmie National Forest** (part of 279 reintroduced statewide between 2008 and 2021 by WDFW, the National Park Service, Conservation Northwest, and partners). The **Nooksack Tribe** and other Coast Salish tribes attended release events and offered blessings and songs to welcome the species back to their homelands. Fishers favor mature/old-growth forest with abundant down wood — exactly the habitat preserved in the corridor's ravines. *(Sources: Conservation NW final report; WDFW — fisher; NPS — Washington fisher restoration.)*

4. Other documented species

Species	Status	Where to look
Marbled murrelet	Threatened (ESA)	Nests in old-growth canopy; flies to ocean at dawn/dusk
Northern spotted owl	Threatened (ESA)	Old-growth stands; designated critical habitat in the watershed
Bull trout	Threatened (ESA)	Cold-water tributaries; now accessing upper Middle Fork post-dam removal
Spring Chinook salmon	Endangered (ESA)	Spawning upstream of former dam site — new since 2020
Mountain goat	—	Alpine/subalpine; Twin Sisters area, above treeline
Black bear	—	Everywhere; especially clearcuts with berry crops
Cougar	—	Throughout; rarely seen
Hoary marmot	—	Talus slopes at elevation; listen for the sharp whistle

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Recreation: Trails, Summits & Waterfalls

> Before planning anything off **Forest Service Road 38**, re-read the ACCESS ALERT: FR 38 was washed out in December 2025, cutting the Elbow Lake and Ridley Creek trailheads with no repair timeline as of mid-2026.

Waterfalls

- **Middle Fork Nooksack Falls** (~81 ft): The best waterfall on the Middle Fork — a powerful plunging drop. Reached by starting up the **Ridley Creek Trail** at the end of Middle Fork Road, then scrambling up the riverbed ~1/2 mile. Getting a distant view is not too hard; getting close requires a sketchy traverse across a very loose hillside. It's glacier-fed by the Deming Glacier, so it runs fullest on hot summer days. *(Source: Aaron's Waterfall World.)*
- **Lower Middle Fork Nooksack Falls** (~43 ft, two tiers): A smaller falls ~500 ft downstream of the main falls, wedged among enormous boulders. You pass right by it on the way to the main falls. *(Source: Northwest Waterfall Survey.)*

Trails

- **Elbow Lake Trail #697:** From the FR 38 (north) trailhead, the trail crosses the Middle Fork and climbs steeply through old-growth to a Twin Sisters viewpoint (~2 mi), then eases past boggy areas to Elbow Lake (~3.7 mi) and smaller Lake Doreen. Magnificent ancient Alaska yellow-cedars; popular fishing. A southern trailhead is reachable from Forest Service Road

12 (Pioneer Horse Camp) via Bell Creek (a ford). *(Sources: WTA — Elbow Lake; go-washington #697.)*

- **Ridley Creek Trail to Mazama Park / Bell Pass:** From the end of Middle Fork Road, drop in and ford the Middle Fork (the trickiest part of the day), then climb east in shade through hemlock and yellow-cedar, with steep dry-streambed crossings (including two branches of Ridley Creek). At ~3 miles the terrain opens into Mazama Park meadow with views of the Twin Sisters and Baker's glaciated SW face. Mazama Park has a three-sided shelter and connects toward Bell Pass, Elbow Lake, and (via the Mt. Baker NRA) **Park Butte lookout**. Note the trail's hidden outlet at the river for the return, and remember the ford rises through a warm day. *(Source: Lynden Tribune feature.)*

Twin Sisters scrambles & climbs

The Twin Sisters Range offers some of the finest scrambling in Washington on **grippy dunite** — coarse, olivine-rich rock that feels like 50-grit sandpaper under your hands. Approaches are long logging-road grinds, usually via **Dailey Prairie**; most parties **bike the approach** and coast out.

- **North Twin Sister, West Ridge** (6,640 ft): The classic — roughly 1,500 ft of sustained **Class 3-4** scrambling on solid rock. Route-finding is notoriously tricky around the obelisk tower; stray right and you'll find harder Class 4-5 terrain. If you're on 5th-class rock, you're off route. Common descent is the North Couloir (glissade where safe in season). Roughly 16 miles round trip / ~5,400-5,500 ft gain from the car. *(Sources: The Mountaineers; SummitPost — North Twin; Steph Abegg trip report.)*
- **South Twin Sister** (~7,000 ft): The range high point; objectives include the Green Creek Arete and other alpine rock up to ~5.6.
- **The range** also holds Skookum Peak, Hayden Peak, Little Sister, Cinderella Peak, Shirley Peak, and more — a peakbagger's playground. *(Source: SummitPost — Twin Sisters Range.)*

> **Access etiquette:** The logging roads and surrounding land are private and gated. Relations between climbers, timber operators, and Olivine Corporation have historically been touchy. Don't block roads, respect closures, and don't give anyone a reason to lock climbers out.

Winter: backcountry skiing & splitboarding

The Twin Sisters are the closest "real mountains" to Bellingham, and they catch Pacific storms head-on — steep couloirs and big alpine faces above bowls and glades. Winter access is committing: gated roads, long approaches (often via Dailey Prairie, partly on bike or skin), and serious avalanche terrain. The documented **North-to-South Twin ski traverse** is roughly **35 miles with ~12,000 ft of gain**, skiing from both summits. *(Sources: Turns All Year — Twin Sisters traverse; WildSnow — North Twin; Baker Mountain Guides.)*

The Mt. Baker Marathon (1911-1913) — local history you can still trace

The corridor's most famous race wasn't a trail race as we'd know it. The **Mt. Baker Marathon** (1911-1913) was a ~120-mile round trip: competitors traveled from Bellingham to the mountain **by car or train**, ran to the summit of Baker and back down, then returned to town. **Joe Galbraith** — an Acme logger/rancher, and son of murdered Justice of the Peace A.A. Galbraith — won the **inaugural 1911 race in 12 hours 28 minutes**, ascending via the **Deming Trail**. He was forced out of the 1912 race by an automobile accident, and the event was discontinued after 1913 over safety concerns (crevasse falls and a near-disaster). The Deming Trail he ran climbs out of this watershed. *(Sources: HistoryLink — Joe Galbraith wins 1911; Whatcom Museum exhibit; Ultrarunning History.)*

Live Conditions: Sensor & Camera Network

Beyond the rocks and history, the corridor now has a small **environmental sensor and camera network** quietly running in the woods — a personal project to get real "eyes on conditions" up high, because valley and airport forecasts rarely match what the ridges are actually doing. If you find a small QR-tagged box or a camera out here, this is what it is.

What's out there:

- **Trail cameras** watching terrain and conditions (not people) — commercial cellular cams (Vosker, GardePro) feeding a project website, plus custom high-resolution "Woods Cam" builds in testing. Imagery filtering drops frames with people before anything is published; no faces are posted.
- **Weather & air-quality nodes** measuring temperature, humidity, pressure, particulate/AQI, snow depth, soil temperature, and wind. Built on rugged off-the-shelf Dragino sensors and custom Particle (Muon/Boron) hardware.
- **Ultrasonic wind sensors** aimed at **differential wind** between sites — e.g., the NW face of North Twin vs. South Twin vs. Bowman — kept around 3,500-4,000 ft where sun cycles may limit rime icing.
- **A LoRaWAN mesh** with a solar-powered, cellular-backhaul gateway on **North Twin** (heated-battery upgrades underway after cold-snap shutdowns), tying remote nodes back to a home backend (The Things Network / Particle Cloud -> MQTT -> Node-RED -> TimescaleDB).

How to view conditions:

- **Most live weather is published to Windy.com** (temperature, pressure, wind, etc.) — look for the corridor's stations.
- **Air quality, soil/probe temperatures, and some specialty channels** are recorded but not yet on a public site. Ideas for a good public home for those streams are welcome.
- **Trail-camera images** live on the project website (thousands of frames; a seasonal timelapse is in the works).

Sites (approximate): Bowman Ridge / Bowman Mountain, North Twin (gateway + approach), South Twin approach, Dailey Prairie, Baker View Turn, West Ridge, and several Mosquito Lake Road-area nodes.

> **If you find a tagged box or a camera:** It's a conditions/safety project, not surveillance. Scan the QR tag to land on the project site, or reach out there with any concerns — placement can be adjusted or gear pulled. Cameras are being relocated higher and farther from travel corridors where views stay useful.
>
> *(See the companion write-up `TWINS-WEATHER-AND-SENSORS-NARRATIVE.md` for the full story and architecture. TODO: add the public project-website URL and direct Windy station links once finalized.)*

Timeline: This Land Through Western Contact

What you see driving these roads is the layered result of ~170 years of western activity on top of thousands of years of Nooksack presence. Rows are in chronological order; where an entry spans years, it is placed by its start date.

Date	What happened	What you can still see
Time immemorial	Nooksack people (Lhechalosem) inhabit the *Nuxwt'iqw'em* (Middle Fork) watershed. Seasonal round: winter villages at low elevation, summer camps at high elevation for hunting, gathering, and spirit questing.	Culturally modified cedar trees (bark-stripped), lithic scatters, fire-cracked rock, cairns, trail routes. The TCP designation protects this continuing use.
~6,500 years ago	Largest lahar ever recorded from Mount Baker flows down the Middle Fork valley.	Lahar deposits visible in riverbank terrace cuts (see Geology).
1855	Treaty of Point Elliott signed at Mukilteo. The Nooksack did not attend — their rights were signed away by Lummi Chief Chow-its-hoot. Treaty cedes land but	Treaty rights remain legally enforceable today (Boldt Decision, 1974).

Date	What happened	What you can still see
	guarantees fishing, hunting, and gathering rights.	
1858	Fraser River Gold Rush. Thousands of miners pass through Whatcom County headed to British Columbia. Few stay, but trails are cut and the Nooksack valley is mapped.	The "Nooksack Road" trail to Canada was cut during this period.
1870s-1880s	First permanent Euro-American homesteaders arrive in the lowlands: Van Zandt (1883), Galbraith (~1884), others along the river. Most settlement is on the valley floor — Deming, Acme, Van Zandt.	Place names. McBride homestead on the lower Middle Fork floodplain (fruit trees, foundation stones).
1884	Indian Homestead Act. Many Nooksack people, denied a reservation, file individual homestead claims on their traditional village sites.	Nooksack community centered at Deming remains to this day.
1885-1900	GLO surveys of the Middle Fork and tributaries. Upper watershed remains unpatented — too steep, too remote for farming.	Survey markers and section corners still reference these original plats.
1891	Bellingham Bay & British Columbia Railroad reaches Sumas. Rail access transforms the timber economy.	Railroad grades converted to roads.
1897	President Cleveland proclaims the Washington Forest Reserve, withdrawing the unclaimed high country from future entry.	The "federal line" east of the corridor (now Mt. Baker-Snoqualmie NF). See Why the East Side Is Federal.
1898	Lake Whatcom Logging Company formed (Bloedel, Donovan, Larson).	Beginning of industrial-scale logging in the county.
1900	Frederick Weyerhaeuser buys 900,000 acres of Northern Pacific grant land in Washington for $6/acre.	The checkerboard ownership pattern visible from space today.
1900s-1910s	First-generation hand logging in the Middle Fork corridor. Crosscut saws, springboards, steam donkeys, ox teams, skid roads. Saxon logging camp established ~1905.	Springboard-notch stumps (6-10 ft diameter), steel cable, choker hardware, skid-road depressions.
1911	A.A. Galbraith and Winn Stevens killed in the Saxon-camp/Acme shootout. Joe Galbraith wins the first Mt. Baker Marathon (12:28).	Galbraith name on mountain, creek, and chromite claims.

Date	What happened	What you can still see
1911-1913	Mt. Baker Marathon run for three years, then discontinued over safety.	The Deming Trail route out of this watershed.
1920-1937	Bloedel-Donovan builds 50+ miles of logging railroad through the Saxon-South Fork corridor. Peak of railroad logging.	Railroad spikes, tie plates, bridge abutments, trestle pilings (visible at low water).
1926-1934	Mineral prospectors stake chromite, olivine, and other claims on Twin Sisters. Lambert brothers (1926), Washington Chrome Co. (1931), Galbraith Group (1934).	Ribbon Mine open cut, Danny Mine cliff workings, claim markers.
1927	Deming Glacier jokulhlaup (glacial outburst flood) sends house-sized ice and boulders down the Middle Fork.	Garage-sized boulders on river terraces; 1927-dated trees on the low terrace.
1927-1941	Longview Fibre Company (founded 1927, Longview WA) begins in 1941 buying its own timberland to secure wood supply — eventually 570,000+ acres over 60 years.	How much of the corridor's timberland passed from scattered small owners into one corporate portfolio.
~1930s-1960s	"Old Man Dailey" — a solitary squatter — lives in a small cabin at the edge of the sphagnum prairie (~3,200 ft) between Bowman and Twin Sisters. No formal claim; known only through logger oral tradition. Dave Tucker photographed the cabin still standing in 1990; it was later burned down (reportedly by federal authorities).	The cabin is gone. The prairie — and the name — remain. Now a 229-acre DNR Natural Area Preserve.
1943	USGS investigates Twin Sisters chromite for WWII strategic minerals; concludes deposits are too small for commercial mining.	USGS Open-File Report 43-39. Prospect pits and trenches on the upper mountain.
1944-1946	Bloedel-Donovan Lumber Mills liquidates. J.H. Bloedel donates 12.5 acres of the Lake Whatcom mill tract to Bellingham (now Bloedel Donovan Park). Cutover Whatcom land is sold piecemeal.	The end of the railroad logging era in Whatcom County.
1946	Olivine mining begins at the Swen Larsen Quarry site on NW Twin Sisters.	Access road and quarry visible today (now Olivine Corporation, active since 1963).
1949	Nooksack Lookout (Bowman Mountain Lookout) built at 3,350 ft — 80-foot pole tower with L-6 cab.	Tower demolished ~1968. Possible footings and cleared ground at summit.

Date	What happened	What you can still see
1961	City of Bellingham builds 24-foot diversion dam on the Middle Fork. No fish passage. Blocks 16 miles of spawning habitat.	Dam removed 2020; channel restoration completed 2022.
1963	Georgia-Pacific acquires Puget Sound Pulp & Timber Co. and continues managing Whatcom County timberlands for its Bellingham pulp mill.	Some of these lands later consolidate into the Longview Fibre portfolio.
1968	Nooksack Lookout tower and ground cabin destroyed.	Footings/bolts/cleared ground at the Bowman summit.
1973	Nooksack Tribe gains federal recognition after being unrecognized since 1855.	The modern tribal government and Cultural Resources program.
1973	Mt. Baker and Snoqualmie National Forests merged into the Mt. Baker-Snoqualmie National Forest.	The federal land east of the corridor.
1974	*U.S. v. Washington* (Boldt Decision) reaffirms treaty fishing rights for Nooksack and Lummi.	Legal foundation for tribal co-management of salmon and rivers.
1984	Washington State Wilderness Act designates the Mount Baker Wilderness (117,528 acres), including the Twin Sisters peaks.	No roads, no logging, no motorized use in the wilderness.
2007	Brookfield Asset Management (Toronto) buys Longview Fibre; splits it, and the 588,000 acres of timberland become Longview Timber LLC (a pure investment vehicle).	Private-equity ownership of the forests.
2013	Weyerhaeuser buys Longview Timber LLC from Brookfield for $2.65B (645,000 acres across WA and OR), increasing its PNW holdings by ~33%.	The corridor timberlands now owned by the company Frederick Weyerhaeuser founded in 1900.
2013	May 31 debris flow descends the Middle Fork (detected on PNSN seismometers).	New boulder levee ~15 ft above the channel near the Ridley Creek trail.
2020	Middle Fork diversion dam removed (City of Bellingham, Nooksack Tribe, Lummi Nation, American Rivers, NOAA, WDFW, Paul G. Allen Family Foundation).	Former dam site visible. Spring Chinook, steelhead, bull trout now upstream.
2021 (Apr-Jul)	Weyerhaeuser sells 145,000 acres in the North Cascades to Hampton Lumber for	Hampton-managed timberland — the land you drive through.

Date	What happened	What you can still see
	$266M (highest operating cost, lowest productivity acreage). Covers six counties.	
2021 (Oct)	Hampton sells 50,747 eastside acres (Chelan, Kittitas, Snohomish, King) to Chinook Forest Partners; keeps the Whatcom/Skagit core.	Current owner of the Middle Fork corridor. Recreation by permit only (~600 motorized, ~150 non-motorized/year).
2022	Dam-removal channel restoration complete; effectiveness monitoring runs through 2030.	Free-flowing river; ongoing monitoring confirms fish passage.
2018-2020	Cascades Fisher Reintroduction: 89 fishers from Alberta released across the North Cascades / Mt. Baker-Snoqualmie NF; Nooksack Tribe offers blessings.	A recovering forest carnivore in the corridor's old-growth.
Dec 2025	Floods wash out FR 38 (Middle Fork Road), cutting the Elbow Lake and Ridley Creek trailheads.	Damaged road/trail access; repairs planned, no timeline as of mid-2026.
Present	Hampton contracts logging to firms like A.L.R.T. Corporation (Everson). DNR and USFS manage interspersed public land. Nooksack Tribe continues traditional use under treaty rights and TCP protections.	Active logging, active quarrying, active cultural use, active conservation — all overlapping on the same roads.

How 145,000 Acres Ended Up in One Company's Hands

The story of who owns this land is really the story of how the Pacific Northwest timber economy consolidates.

Phase 1 - Railroad land grants (1864-1900). In 1864, Congress chartered the Northern Pacific Railroad and paid for its construction not with cash but with land — every odd-numbered section (a section is one square mile, 640 acres) in a corridor 40 miles wide (20 miles on each side of the tracks) through the territories. That's every other square mile in a band stretching from Lake Superior to Puget Sound. The even-numbered sections stayed federal. This created a literal checkerboard visible on any ownership map — alternating squares of railroad land and government land.

Why alternating? Every township (6 x 6 miles) in the West was divided into 36 sections, each one square mile (640 acres), numbered in a snake pattern:

```
 6 |  5 |  4 |  3 |  2 |  1
 7 |  8 |  9 | 10 | 11 | 12
18 | 17 | 16 | 15 | 14 | 13
19 | 20 | 21 | 22 | 23 | 24
30 | 29 | 28 | 27 | 26 | 25
31 | 32 | 33 | 34 | 35 | 36
```

Congress gave the railroad all odd-numbered sections (1, 3, 5, 7...) within the corridor. The government kept all even-numbered sections (2, 4, 6, 8...). Color in the odds and leave the evens blank and you get a literal checkerboard. The railroad also got a narrow continuous right-of-way strip (200-400 ft wide) for the actual tracks — that's where the railroad was built. The checkerboard of sections spreading 20 miles on each side was payment, not the route.

The theory: (1) the government keeps half the land and doubles the price of its retained sections, because land near a railroad is worth more; (2) the railroad sells its sections to settlers to fund construction. In practice, the railroads discovered the timber was worth far more than farmland, held the sections, and sold them in bulk to timber companies. The government's retained even-numbered sections became national forest — and the checkerboard fragmentation became a permanent headache. You can see the pattern from space today: alternating clearcut and forested squares where private timber companies and the Forest Service manage side-by-side with completely different objectives.

The Northern Pacific grant was enormous — roughly 40 million acres across the West. In 1900, **Frederick Weyerhaeuser bought 900,000 acres** of Northern Pacific grant land in Washington for $6 per acre ($5.4 million total) — one of the largest private land sales in American history. Weyerhaeuser said: "This is not for us, or for our children, but for our grandchildren." Twelve years later someone calculated he had paid about 10 cents per 1,000 board feet.

Phase 2 - Local logging empires (1900-1945). Companies like Bloedel-Donovan (Bellingham) and Puget Sound Pulp & Timber assembled their own timberlands through direct purchase, timber sales, and railroad access. They built mills, railroads, and company towns. When the old growth ran out or the Depression hit, they liquidated. Bloedel-Donovan shut down in 1944-46.

Phase 3 - Paper companies buy the land (1941-2000). Longview Fibre Company, a Longview WA paper mill that had been buying waste wood chips from sawmills, realized it needed its own timber supply. Starting in 1941, it spent 60 years buying small parcels — the leftover pieces from liquidated logging companies, small landowners, and cutover tracts — and assembled 570,000+ acres across Washington and Oregon. This patient, decades-long accumulation is how the Middle Fork corridor timberland ended up in one portfolio. Georgia-Pacific (which swallowed Puget Sound Pulp & Timber in 1963) held other pieces; some of those eventually consolidated into the same portfolio.

Phase 4 - Private equity (2007). Brookfield Asset Management (Toronto) bought Longview Fibre, split the paper mill from the timberland, and created "Longview Timber LLC" — a pure timberland investment vehicle. The forests were now a financial asset managed for return, not for feeding a local paper mill.

Phase 5 - Weyerhaeuser absorbs everything (2013). Weyerhaeuser bought Longview Timber for $2.65 billion. This brought lands that Longview Fibre had spent 60 years assembling under the same corporate roof as the 1900 Northern Pacific grant lands.

Phase 6 - Weyerhaeuser prunes (2021). Weyerhaeuser decided the North Cascades lands were their worst-performing western acreage — high elevation, high operating cost, low Douglas-fir content, no connection to any Weyerhaeuser mill. They sold 145,000 acres to Hampton Lumber for $266M. Hampton wanted them to feed its Darrington sawmill. Hampton then flipped the eastside acres (Chelan, Kittitas, part of Snohomish/King) to Chinook Forest Partners and kept the Whatcom/Skagit core — the land you drive through.

The result: land that was Nooksack territory for millennia, then a checkerboard of railroad grants and homestead claims, then logged by Bloedel-Donovan and others, then patiently assembled parcel by parcel by a paper company over 60 years, then bought by a Canadian private-equity fund, then absorbed by the largest private timberland owner in America, then sold to a family-owned Oregon lumber company — is now Hampton Lumber land, managed for sustained-yield timber harvest feeding a sawmill 90 miles south.

Why the East Side Is Federal Land

If you look at a land-ownership map of this area, there's a sharp line where Hampton's private timberland ends and the Mount Baker-Snoqualmie National Forest begins. The Twin Sisters peaks, the alpine zones, and everything east of the main ridge is federal. Why?

The short answer: That land was never claimed by anyone, so the federal government kept it.

The longer answer: All land in Washington started as federal public domain — owned by the United States after the Oregon Treaty with Britain (1846). The government's plan was to give it away: to railroads (Northern Pacific land grants), to settlers (Homestead Act, 1862), to the state (school trust sections), and to miners (General Mining Law, 1872). Anything nobody claimed stayed federal.

The lower-elevation, timberable land along the Middle Fork corridor was attractive — settlers could farm the floodplain, loggers could reach the timber, railroad grants conveyed the odd-numbered sections. But the high-elevation, steep, rocky, glaciated terrain of the Twin Sisters and the upper Mount Baker slopes? Nobody wanted to homestead at 5,000 ft on bare dunite.

Nobody filed a timber claim on a cliff face. The GLO surveyors mapped it, but no one showed up at the land office to patent it.

So when President Grover Cleveland proclaimed the Washington Forest Reserve on February 22, 1897 — one of his last acts in office, under the Forest Reserve Act of 1891 — those unclaimed high-elevation sections were still public domain. Cleveland's proclamation simply withdrew them from future entry. The land went from "federal land nobody has claimed yet" to "federal land nobody can ever claim." That's how national forests are made — not by the government taking land, but by the government keeping land it already owned.

The timeline:

- **1891:** Forest Reserve Act gives the president power to set aside public domain as forest reserves.
- **1897:** Cleveland proclaims the Washington Forest Reserve (northern Cascades) and Mt. Rainier Forest Reserve (southern Cascades). Eight million acres total.
- **1905:** Forest reserves transferred to the new U.S. Forest Service under the Dept. of Agriculture.
- **1908:** Washington Forest Reserve split into the Washington National Forest (Canada to Skagit River) and Snoqualmie National Forest (Skagit to Green River).
- **1924:** Washington National Forest renamed Mt. Baker National Forest.
- **1973:** Mt. Baker and Snoqualmie merged into the Mount Baker-Snoqualmie National Forest.
- **1984:** Washington State Wilderness Act designates the Mount Baker Wilderness (117,528 acres), which includes the Twin Sisters peaks — no roads, no logging, no motorized use, ever.

Why DNR parcels are mixed in: When Washington became a state in 1889, the federal Enabling Act granted sections 16 and 36 of every township to the state to fund public schools. These are trust lands — the DNR is legally required to manage them to generate revenue for schools and counties. That's why you see DNR timber sales and state trust parcels interspersed with Hampton land and national forest. They've been there since statehood. Additional state parcels were added in the 1920s-1930s when private timberland owners defaulted on property taxes during the Depression, and counties transferred the forfeited land to the state.

So three legal events created the ownership map you drive through today:

1. **1889 - Statehood.** Sections 16 and 36 go to the state (now DNR trust land).
2. **1897 - Cleveland** reserves the unclaimed high country as national forest.
3. **Everything else** — the lower, reachable, timberable land — was claimed by settlers, granted to the Northern Pacific Railroad, or purchased by timber companies. That's the

private land (now Hampton).

Traditional Cultural Property: What It Means

The entire Middle Fork watershed is designated a **Traditional Cultural Property (TCP)** — a category formally defined in **National Register Bulletin 38 (1990)** under the authority of the **National Historic Preservation Act (1966)**. This means the landscape itself is eligible for / on the National Register of Historic Places because it is essential to the continuing cultural identity of the Nooksack Indian Tribe. It is not a museum — it is a living cultural landscape where traditional practices still happen.

Nooksack and the land (publicly documented):

- **Spirit questing:** Individual isolation, fasting, cold-water bathing in creeks and mountain pools, seeking a guardian spirit. Central to Coast Salish spiritual life for thousands of years. Specific sites are kept private.
- **Hunting, fishing, gathering:** Treaty-guaranteed rights (Point Elliott Treaty, 1855; Boldt Decision, 1974). Salmon, elk, deer, berries, roots, cedar bark.
- **Cedar bark gathering:** Stripping bark from living cedars for basketry, clothing, and other uses. This creates Culturally Modified Trees (CMTs) — old cedars with long, healed-over vertical scars.
- **Traditional religious activities:** The Middle Fork is specifically valued because it is less disturbed than the North or South Forks.

What you might encounter and not recognize:

- **Culturally Modified Trees:** Old-growth cedar with long vertical bark-removal scars. Healed edges. Can be 300+ years old.
- **Lithic scatters:** Small, thin, sharp-edged flakes of dark stone (basalt, chert) on exposed ground — debris from stone tool making/resharpening.
- **Fire-cracked rock:** Clusters of reddened, blackened, or fractured fist-sized rocks on flat ground near water — used for boiling water in wooden containers (Coast Salish peoples did not make pottery).
- **Rock cairns:** Deliberately stacked rocks at elevation — associated with vision quests, burials, trail markers, or game drives.
- **Memorial stones:** Modern painted or inscribed rocks left at meaningful locations (see below).

If you find something: Leave it in place. Photograph it with GPS coordinates. Report it to the Nooksack Tribe Cultural Resources Department or the Washington DAHP. Removing artifacts from state or federal land is illegal (RCW 27.53). The Hampton recreation permit also prohibits disturbing cultural artifacts.

Memorial Stones at Dailey Prairie

At the edge of Dailey Prairie, hand-painted and inscribed rocks have been placed among the moss and bog laurel. These are **modern personal memorial stones** — painted with messages like "Live Life Love," "Mom," "Brother," "Husband," and longer inscriptions about wind, memory, and loss. A small artificial flower is placed with them.

These are grief memorials — someone brought their loved one's memory (or ashes) to a place that mattered to them. This is a common contemporary practice in the Pacific Northwest backcountry, especially at places with personal significance. Given Dailey Prairie's history as a place where people have come for solitude — from Old Man Dailey's cabin, to cross-country skiers, to whoever left these stones — it makes sense that someone chose this spot.

They are not archaeological artifacts. They are not ancient. But they are clearly placed with care and intention. **Respect them as you would any memorial.**

Who Operates Here Now

Company	Role	Notes
Hampton Lumber	Landowner, timber producer	Acquired 145,000 acres from Weyerhaeuser (July 2021). Feeds Darrington sawmill. Hampton foresters plan harvests.
A.L.R.T. Corporation (Everson)	Contract logger for Hampton	~55 employees. Logging, road building, rock crushing, hauling. Most likely the crews you see on Hampton land.
Nielsen Brothers Inc. (Bellingham)	Contract logger, DNR timber-sale bidder	~75 employees. Founded 1979 by David & Robert Nielsen. Bids on DNR state timber sales in Whatcom County. Also operates on own timberland (sold 754 acres of Lake Whatcom watershed to City of Bellingham, Feb 2025). May work DNR parcels interspersed in the corridor.
Olivine Corporation (Bellingham)	Quarry operator	Active olivine quarry on NW Twin Sisters since 1963. Trucks 40 miles to Bellingham grinding plant.

Company	Role	Notes
DNR	State trust-land manager	Manages interspersed state trust parcels; conducts its own timber sales.

Quick-Reference Waypoint Table

Approximate locations for the features in this guide. Coordinates marked *(approx.)* are derived from descriptions, GNIS, or map references and should be field-verified before relying on them.

Feature	Approx. location	Access notes
Middle Fork lahar / gravel bars	End of Middle Fork Road, near old dam site	Walk to gravel bars; Ridley Creek Trail start
Middle Fork Nooksack Falls (~81 ft)	~1/2 mi up the riverbed from Ridley Creek Trail	Scramble; loose hillside for close views
Lower Middle Fork Falls (~43 ft)	~500 ft downstream of main falls	Passed en route to main falls
Twin Sisters dunite (viewpoint)	Peaks to the SE/E from the corridor	Visible from many road points
North Twin Sister (6,640 ft)	Twin Sisters Range	West Ridge Class 3-4; approach via Dailey Prairie (bike)
South Twin Sister (~7,000 ft)	Twin Sisters Range (high point)	Alpine rock; approach via Dailey Prairie
Clay Banks Landslide	~1.8 mi upstream of Baker Hwy bridge	View from accessible vantage points
Van Zandt Dike	E side of Highway 9, Deming-Acme	Roadside view; labeled from Nooksack Lookout
Former diversion dam site	Upper Middle Fork corridor	Restored channel; fish-screened intake nearby
Olivine Corporation quarry	SE1/4 NE1/4 Sec 34, T38N R6E *(approx.)*	~11 mi E via old Nooksack Lookout road; **active — do not enter**
Asbestos veinlets, Bowman	NW1/4 Sec 29, T38N R6E *(approx.)*	Bowman Hill logging area; **observe only**
Nooksack Lookout site	Bowman Mtn summit, 3,350 ft	Footings/cleared ground remain
Marble Peaks / Boulder Creek quarry	W1/2 Sec 22, T40N R6E *(approx.)*	~3 mi NE of Maple Falls; last mile of road abandoned

Feature	Approx. location	Access notes
Doaks Creek quarry	Center Sec 19, T40N R6E *(approx.)*	~1.25 mi N of Maple Falls
Dailey Prairie NAP	~3,200 ft pass, Bowman-Twin Sisters	229-acre DNR preserve; memorial stones at edge
Orsino Creek	N48.69039 W122.04071 (GNIS)	Drains S from Dailey Prairie pass
Seymour Creek	N48.75762 W122.00987 (GNIS)	Near Twin Sisters
Mosquito Lake	48.7691 N, 122.1205 W	Main access corridor from lowlands

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Glossary

- **Dunite:** A coarse-grained ultramafic rock made almost entirely of olivine; mantle rock. Fresh dunite is green; it weathers reddish-brown. The Twin Sisters body is the largest exposed dunite in the contiguous US, and its grippy texture makes for excellent scrambling.
- **Olivine:** The green magnesium-iron silicate mineral that makes up dunite; gem-quality olivine is peridot. Quarried locally for foundry sand and refractory uses.
- **Chromite:** A black chromium-iron oxide mineral occurring in folded bands within the dunite; the target of 1930s-40s claims.
- **Serpentinite / serpentinized peridotite:** Ultramafic rock altered by water; green-black, can host silky white chrysotile (asbestos) veinlets. Do not crush — fibers go airborne.
- **Chrysotile:** White, silky "cross-fiber" serpentine asbestos found in thin veinlets on Bowman Mountain.
- **Lahar:** A volcanic mudflow/debris flow — a fast slurry of water, rock, mud, and ice off a volcano. The ~6,500-year-old Middle Fork lahar is the largest known from Mount Baker.
- **Jokulhlaup (glacial outburst flood):** A sudden release of water (and debris) from a glacier; the Icelandic term. The June 1927 Deming Glacier event left house- and garage-sized boulders on the Middle Fork terraces.
- **Shermanite:** Informal name for hydrothermally altered, bleached andesite from Mount Baker's Sherman Crater, carried downriver by lahars and floods; found as crumbly white-yellow cobbles in the gravel.
- **TCP (Traditional Cultural Property):** A place eligible for the National Register because of its role in a living community's cultural identity (National Register Bulletin 38, 1990). The whole Middle Fork watershed is a Nooksack TCP.
- **CMT (Culturally Modified Tree):** A living tree (usually old-growth cedar) bearing healed

scars from traditional bark or plank harvesting; can be centuries old.

- **GLO (General Land Office):** The federal agency that originally surveyed and patented public-domain land; its 1880s-1900s plats still define section corners here.
- **Checkerboard / section:** A section is one square mile (640 acres); a township is 36 sections. The Northern Pacific railroad grant took every odd-numbered section, producing the alternating private/public "checkerboard" you can see from the air.
- **Springboard notch:** A rectangular slot cut into a big stump where loggers wedged a plank to stand above the thick butt swell while hand-sawing.

References

- **DNR Bulletin 57:** *Mines and Mineral Deposits of Whatcom County, Washington* (Wayne S. Moen, 1969) — dnr.wa.gov (free PDF; includes fold-out maps of mine locations and claims)
- **GSA Field Guide 38** (Tucker et al., 2014): "Mount Baker lahars and debris flows" — the road-log field-trip guide for Middle Fork lahar stops
- **MBVRC:** Mount Baker Volcano Research Center — mbvrc.wordpress.com (field-trip reports with photos)
- **NW Geology Field Trips:** Ridley Creek Trail geology guide (Middle Fork lahar, 1927 jokulhlaup, 2013 debris flow)
- **DNR WA100 Geotourism:** wa100.dnr.wa.gov/north-cascades/twin-sisters
- **Richardson & Galloway (2011):** *Nooksack Place Names* (UBC Press)
- **Galloway & Richardson (1983):** "Nooksack Place Names: An Ethnohistorical and Linguistic Approach" — lingpapers.sites.olt.ubc.ca
- **City of Bellingham dam removal:** cob.org/services/environment/restoration/middlefork
- **American Rivers — Middle Fork Nooksack:** americanrivers.org
- **USGS Clay Banks monitoring:** usgs.gov/programs/landslide-hazards
- **Hampton Recreation Program:** hamptonlumber.com/recreation — permits via hampton.myoutdooragent.com
- **The Diggings:** thediggings.com — interactive map of BLM mining claims (search Olivine Corporation, Ribbon Mine, Danny Mine)
- **Mindat.org:** Twin Sisters minerals, Ribbon Mine, Danny Mine
- **Stilson, Meatte & Whitlam:** *A Field Guide to Washington State Archaeology* (DAHP/WSDOT/DNR, rev. 2003) — dahp.wa.gov (free PDF)
- **Nooksack Tribe Cultural Resources:** nooksacktribe.org/departments/cultural-resources — contact for reporting finds or cultural-resource questions
- **Mt. Baker Marathon:** HistoryLink (1911) · Whatcom Museum exhibit · Ultrarunning History

- **1911 Saxon-camp murders:** HistoryLink — Mike Donnelly · WhatcomTalk — "What's in a Name?"
- **Twin Sisters climbing:** The Mountaineers · SummitPost — North Twin · SummitPost — Twin Sisters Range · Steph Abegg
- **Waterfalls:** Aaron's Waterfall World · Northwest Waterfall Survey
- **Trails:** WTA — Elbow Lake · Lynden Tribune — Middle Fork / Park Butte
- **Forest-road flood damage (Dec 2025):** Cascadia Daily News
- **Fisher reintroduction:** Conservation NW final report · WDFW · NPS
- **Longview Fibre history:** referenceforbusiness.com
- **Weyerhaeuser / Longview Timber (2013):** investor.weyerhaeuser.com
- **Hampton / Weyerhaeuser sale (2021):** hamptonlumber.com
- **HistoryLink — Frederick Weyerhaeuser (1900):** historylink.org/file/5241
- **Acme & Mosquito Lake names:** Acme, WA (Wikipedia) · Revisiting Washington · Mosquito Lake (Wikipedia)

Compiled April 2026; expanded 2026. The Middle Fork watershed is a Traditional Cultural Property of the Nooksack Indian Tribe — treat cultural sites with the respect they deserve. Do not disturb cultural artifacts (also a condition of the Hampton recreation permit). If you find something archaeological, leave it in place, photograph it with GPS, and report it to the Nooksack Tribe or DAHP. Stay out of active logging operations and the Olivine quarry. Old mine workings are dangerous — observe from a distance.